

# Certificate in Accounting and Finance Stage Examination

## Business Insights and Analysis

### Model paper

100 marks



#### **Instructions to examinees:**

- (i) Answer all **NINE** questions.
- (ii) The overall duration of the exam is 3 hours & 15 minutes, which includes the 15-minute reading time.
- (iii) Answer in **black** pen only.
- (iv) Answer Multiple-Choice Questions in the answer script only.

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### **SECTION A (Ranging from 45 to 55 marks)**

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#### **QUESTION 1**

Select the most appropriate answer from the options provided for each of the following Multiple-Choice Questions (MCQs). Each MCQ carries **ONE** mark.

- (i) Which of the following is the primary purpose of conducting a resource audit in an organization?
  - (a) To comply with legal requirements for annual financial audits
  - (b) To identify and assess all significant resources and their effective utilization
  - (c) To determine the organization's market position and level of competition
  - (d) To forecast economic trends and prepare financial statements
- (ii) Which of the following is **NOT** a likely consequence of a poor law and order situation on businesses?
  - (a) Increased employee absenteeism
  - (b) Lower security costs
  - (c) Higher insurance premiums
  - (d) Reduced investment
- (iii) Companies arrange credit lines and overdraft facilities to manage:
  - (a) the credit risk
  - (b) the liquidity risk
  - (c) the market risk
  - (d) the speculative risk
- (iv) A company has recently paid a dividend of Rs. 6 per share, which is expected to grow by 9% per annum in the foreseeable future. The shareholders require an annual return of 19% and the next annual dividend will be paid in one year's time. What would be the market value of each share?
  - (a) Rs. 79.3
  - (b) Rs. 66.7
  - (c) Rs. 65.4
  - (d) Rs. 60.0
- (v) Which **TWO** of the following are examples of strategic or enterprise (speculative) risks?
  - (a) Buying earthquake insurance for a property located in a seismic zone
  - (b) Investing in a start-up company
  - (c) Taking out a mortgage on a house
  - (d) Developing a new product
- (vi) Which of the following best explains a deep-discounted bond?
  - (a) A bond with a low coupon rate that adjusts annually
  - (b) A bond with a low coupon rate that is sold at a significant discount to its fair value
  - (c) A bond with a high coupon rate but sold at a significant discount to its fair value
  - (d) A bond that is available in the market at a significant discount to its issue price

- (vii) Which of the following is correct regarding a bonus issue and a rights issue?
- A bonus issue involves the issuing of shares to existing shareholders, while a rights issue involves offering shares to the general public
  - A bonus issue increases the company's assets, while a rights issue increases the number of shares in circulation
  - A bonus issue increases the number of shares in circulation, while a rights issue increases the company's assets
  - A bonus issue requires the purchase of additional shares at a discount, while rights shares are given at a premium
- (viii) The higher management is considering to implement a structured risk management process in the organization. Which of the following should be considered the most fundamental step?
- Procuring tools for risk evaluation
  - Establishing a dedicated risk management department
  - Formulating a response to potential risk
  - Setting clear expectations about the importance of risk awareness and open dialogue
- (ix) Nowadays, consumers increasingly rely on ratings and reviews from previous buyers, especially through online means, when making purchasing decisions. This is an example of which social factor:
- Attitudes and lifestyles
  - Demography
  - Law and order
  - Health
- (x) Which of the following is the difference between Ijarah and conventional finance lease?
- In Ijarah, the lessor bears the risk of the asset, while in a conventional lease, the lessee bears the risk of the asset
  - In Ijarah, the lessor transfers the ownership, while in a conventional lease, the lessor retains the ownership
  - In Ijarah, any surcharge for delayed payments is credited to income, while in a conventional lease, any such surcharge cannot be credited to income
  - In Ijarah, the rental amount can be variable, while in a conventional lease, the rental amount is fixed

## QUESTION 2

Khalid Masroor (Khalid), an entrepreneur with a strong background in engineering and business, is keen to enter the Electric Vehicle (EV) industry. Although the EV industry is rapidly growing, Khalid is aware of the increasing competition, with established companies leading the industry. To assess the potential of various segments in the EV industry, Khalid, with his risk-averse nature, conducts a Porter's Five Forces analysis and constructs the following table:

| Segment                    | Force 1                | Force 2               | Force 3                       | Force 4                    | Force 5             |
|----------------------------|------------------------|-----------------------|-------------------------------|----------------------------|---------------------|
|                            | Threat of new entrants | Threat of substitutes | Bargaining power of suppliers | Bargaining power of buyers | Competitive rivalry |
| Charging Infrastructure    | Low                    | Low                   | Low                           | High                       | High                |
| Battery Recycling          | High                   | Low                   | Low                           | Low                        | Low                 |
| Specialized Commercial EVs | High                   | High                  | Low                           | High                       | High                |
| Affordable Personal EVs    | High                   | High                  | High                          | High                       | High                |

### Required:

- Advise which segment presents the most viable business opportunity for Khalid. Also, provide reasons for your answer. **(03 marks)**
- For each force, list **three** factors that influence its strength. Also, explain how each of these factors might have impacted the strength of that particular force in your recommended segment in (a). **(11 marks)**

**QUESTION 3**

Northern Hotels Limited (NHL) is planning to build a new resort alongside a lake in Baboon Valley (Valley), a remote area of Pakistan. Although the Valley is known for its scenic views, it is largely cut off from other areas of Pakistan due to its poor infrastructure. There is no sewerage system in the Valley, and all the bio-waste created by the locals is put to organic use. People residing in the Valley have been living below the poverty line.

NHL has obtained all the relevant government approvals. As part of the approval, NHL will build proper road infrastructure leading to the resort. NHL has also been able to enter a deal for the land at a significantly lower market price.

NHL expects to achieve its objective of high profits due to very low operational costs, as the labour is available in the Valley at a significantly reduced rate. On the other hand, the locals have also foresighted that this will bring tourism to the area and will create greater earning opportunities.

**Required:**

Apply Tucker's 5-question model to the above scenario.

**(06 marks)****QUESTION 4**

Ali, a marketing manager at a software company specializing in mobile app development, has the task of overseeing the marketing strategy for IQRA, an Arabic language learning app targeting the Indo-Pak region.

Despite IQRA's unique features, user-friendly interface, and cultural relevance, the fiercely competitive market and limited brand recognition pose significant challenges. Catering to varying language learning perceptions and localization challenges within the region adds complexity to the endeavour. The marketing team must navigate these hurdles with a limited budget while capitalizing on factors like the growing smartphone penetration, increasing interest in Arabic learning, and using social media and influencer marketing. Building partnerships with educational institutions will also be crucial for reaching the target audience effectively. To succeed, Ali's team must closely monitor and adapt to changing user preferences and promptly address any negative reviews or feedback. Moreover, they need to ensure compliance with regional legal and regulatory requirements to maintain user trust.

**Required:**

- (a) Conduct a SWOT analysis for IQRA. **(06 marks)**
- (b) Based on your answer in (a) above, suggest how each identified threat could be tackled using the strengths of IQRA. *(Note: Clearly state the strengths used to answer the question)* **(04 marks)**

**QUESTION 5**

- (a) State briefly the concept of the Triple Bottom Line reporting system. Identify any **two** performance measurement indicators under each aspect of performance discussed in the Triple Bottom Line reporting system. **(05 marks)**
- (b) A fertiliser company is enhancing its sustainability practices and has identified the following Key Performance Indicators (KPIs) to measure its performance:
  - (i) Total water consumption in production per year
  - (ii) Percentage of board seats held by independent directors
  - (iii) Total hours contributed by employees to community and social initiatives
  - (iv) Proportion of energy sourced from renewable resources
  - (v) Number of health and safety incidents reported during the year
  - (vi) Percentage of women in the total workforce and at the executive/management level
  - (vii) Percentage of procurement spending on local suppliers
  - (viii) Average number of employee sick days per year
  - (ix) Ratio of median male salary to median female salary
  - (x) Total number of hours of employee training and professional development per year
  - (xi) Percentage of recycled raw materials used in production
  - (xii) Existence of a published Anti-Corruption Policy

**Required:**

Classify each of the above KPIs under the Environmental, Social, and Governance components of the ESG Framework.

**(06 marks)****SECTION B (Ranging from 45 to 55 marks)****QUESTION 6**

Plasto Plastic Limited (PPL) is an all-equity financed company. The following information has been extracted from PPL's financial statements as at 30 June 2023:

- (i) PPL has ordinary share capital of Rs. 150 million with a face value of Rs. 10 per share.
- (ii) On 30 June 2023, PPL paid an annual dividend of Rs. 25 per share.
- (iii) PPL expects the next three annual dividend payments to be Rs. 25 per share. Thereafter, a 3% per annum growth rate is expected in perpetuity.
- (iv) The equity beta of PPL's share is 0.8, and the average return on the stock market is 21%. The return on government bonds is 12%.

PPL is considering a new investment which would require an initial outlay of Rs. 600 million and is considering financing it with a 1-for-3 rights issue. The following additional information is available regarding the new investment:

- (i) The investment would enable PPL to increase the dividend to Rs. 26 per share on 30 June 2024, and all subsequent dividends would increase by 5% per annum in perpetuity.
- (ii) The investment is riskier than the average of PPL's existing investments and, therefore, the beta would increase to 1.05.

**Required:**

Using the dividend valuation model, determine whether PPL should undertake the new investment.

**(11 marks)****QUESTION 7**

- (a) A corporate treasury plans to invest Rs. 50 million in a 6-month fixed deposit starting in three months. To guard against unfavorable movements in interest rates, it has secured a lender's option with a strike rate of 19.5% on the notional amount. The option expires in three months. The treasury paid an upfront premium of Rs. 250,000, and an additional transaction cost of Rs. 75,000 applies if the option is exercised.

The current and projected KIBOR rates are as follows:

|           | 3-month KIBOR rate | 6-month KIBOR rate |
|-----------|--------------------|--------------------|
| Current   | 18.4%              | 19.8%              |
| Projected | 18.7%              | 18.9%              |

**Required:**

Determine whether the option should be exercised. Also, calculate the net effective interest rate.

**(05 marks)**

- (b) Assume that the date today is **1 March 2025**.

On 1 March 2025, Tech Solutions (TS), a Pakistani technology company, signed an agreement with a German software provider to purchase an AI-powered medical diagnostic system. The payment of EUR 4 million is due on 30 June 2025. To hedge against exchange rate fluctuations, TS is considering a currency option. Each option covers EUR 0.4 million.

The current spot exchange rate is EUR 1 = PKR 290. The following June options are available, each with a premium of PKR 0.6 million per option:

- Call option with a strike price of PKR 295.
- Put option with a strike price of PKR 298.

TS predicts that the spot rate in June could range between PKR 296 and PKR 300 per EUR.

**Required:**Determine whether the option should be purchased. *(Show all necessary computations)* **(05 marks)****QUESTION 8**

Multan Star (MS) is engaged in the manufacturing and selling of a single product, K-100.

The management is in the process of preparing its budgeted profit or loss statement for the year ending 31 August 2023. The following information is available in this respect:

**Information for the year ended 31 August 2022**

(i) Extracts from profit or loss statement:

|                                     | Rs. in '000 |
|-------------------------------------|-------------|
| Sales                               | 90,000      |
| Cost of goods sold:                 |             |
| Material                            | (36,000)    |
| Labour                              | (25,000)    |
| Manufacturing overheads             | (9,000)     |
| Gross profit                        | 20,000      |
| Selling and administration expenses | (2,500)     |
| Profit before tax                   | 17,500      |

- (ii) During the year, MS operated at 80% of its machine production capacity. It manufactured and sold 72,000 units of K-100. Each unit of K-100 requires 3 hours of machine time.
- (iii) 2.5 kg of material is required for the manufacturing of each unit of K-100.
- (iv) Each unit of K-100 requires three labour hours. Labour is hired under a third-party contract according to which MS has to pay for a minimum of 250,000 labour hours. However, each hour exceeding 250,000 hours would be paid at two times of standard labour rate per hour.
- (v) 20% of total manufacturing overheads are fixed. Variable manufacturing overheads are absorbed on the basis of machine hours.
- (vi) All selling and administration expenses are fixed.
- (vii) There are no opening or closing stocks of raw material and finished goods.

**Information and projections for the budget for the year ending 31 August 2023**

- (i) MS would introduce the mini version of K-100 under the name of K-50. The demand for K-50 is expected to be 25,000 units. The selling price of K-50 would be Rs. 750 each.
- (ii) The introduction of K-50 would not affect the existing demand for K-100. In addition, MS has entered into a contract with a new customer for the supply of 10,000 units of K-100 at last year's prevailing price.
- (iii) Any constraint due to production capacity would be met by reducing the production of K-100. However, any shortfall in production of K-100 would be met by purchasing it from the market for Rs. 1,200 per unit.
- (iv) The selling price of K-100 (other than already contracted to supply) would increase by 12%.
- (v) Each unit of K-50 would require 1 kg of material, 1.5 hours of machine and 1.5 hours of labour. The production of K-50 would result in an increase in selling and administration expenses by 20% other than the inflation.
- (vi) All expenses, unless otherwise specified, are subject to an inflation of 5%.

**Required:**

Prepare a budgeted profit or loss statement for the year ending 31 August 2023.

**(12 marks)**

**QUESTION 9**

Zaid Limited (ZL) launched a product called 'Zing' two years ago and prepared a five-year projection for it. In the first year, Zing performed well and achieved all milestones. However, in the second year, the machine used for the production of Zing developed certain operational issues. While ZL was able to maintain the production of the originally estimated quantity, the quality of Zing was significantly affected, resulting in lost sales. ZL is now considering either replacing the machine and continuing with Zing's production or discontinuing Zing's production.

The following information is available in this respect:

**Original estimates:**

- (i) ZL purchased an old machine costing Rs. 120 million for the production of Zing and had to incur an overhaul cost amounting to Rs. 25 million immediately, and another overhaul of the same amount would have to be incurred at the end of the third year.
- (ii) As per ZL's policy, the machine was depreciated at a rate of 25% using the reducing balance method. It was anticipated that the residual value of the machine would be equal to its written-down value at the end of the project, i.e., the fifth year.
- (iii) Initial investment in working capital was assessed at Rs. 20 million, with no further additions required in subsequent years. ZL anticipated that only 25% of this investment would be realized at the end of the fifth year.
- (iv) A warehouse was rented for a 5-year period at an annual rent of Rs. 5 million, subject to an annual increase of 10%. Early termination of the agreement would incur a penalty equivalent to 6 months' rent.
- (v) Production and sales were estimated at 40,000 units per annum for all the years. However, in the second year, only 15,000 units were sold due to quality issues.
- (vi) The sales price for the first year was Rs. 1,500 per unit, with an annual increase of 10%.
- (vii) Variable costs were estimated at Rs. 650 per unit, with fixed costs (other than rent) associated with Zing at Rs. 1.5 million per annum. Both costs were subject to a 7% annual increase.
- (viii) The weighted average cost of capital is 18%.
- (ix) ZL operates in a tax-free environment.
- (x) All cash flows would arise at the end of the year, except where stated otherwise.

**Option 1: Replace the machine and continue with the production of Zing**

- (i) A Chinese supplier has offered ZL the opportunity to trade in the old machine for Rs. 60 million and upgrade to the latest machine for an additional investment of Rs. 100 million. This would help ZL in maintaining the production of Zing at standard quality for the remaining period of the project. At the end of the project, ZL expects selling this upgraded machine with a 15% profit above its written down value.
- (ii) ZL can negotiate a contract with an existing customer to sell the existing stock of Zing at 85% of the price based on the original estimates.
- (iii) The new machine is expected to reduce variable costs by 8% compared to the original estimates.
- (iv) All other information would remain consistent with the original estimates.

**Option 2: Discontinue production of Zing**

- (i) Under this option, the existing stock of Zing would be sold at 80% of the price based on the original estimates.
- (ii) The machine would be sold for Rs. 50 million on an 'as is where is' basis.
- (iii) 35% of the working capital would be realized.
- (iv) All other information would remain consistent with the original estimates.

**Required:**

Evaluate both options by using the net present value method. Recommend the best course of action for ZL to follow. *(Net present value based on original estimates is not required)* **(16 marks)**

**(THE END)**